MILITARY SPECIFICATION

FIBER SHEET, VULCANIZED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers the requirements for vulcanized fiber sheets ranging in thickness from 0.010 to 1.00 inch (0.25 to 25.4 mm). Thickness between 0.020 and 0.125 inch (0.51 and 3.18 mm) shall be trunk and case grade. Other thickness shall be commercial grade (see 6.3).

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 9330

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MILITARY

MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads for 40" x 48" Pallets.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 202 - Sampling and Testing Untreated Paper Used for Electrical Insulation.
D 256 - Impact Resistance of Plastics and Electrical Insulating Materials.
D 570 - Water Absorption of Plastics.
D 618 - Conditioning Plastics and Electrical Insulating Materials for Testing.
D 619 - Vulcanized Fiber Used for Electrical Insulation.
D 785 - Rockwell Hardness of Plastics and Electrical Insulating Materials.
D 790 - Flexural Properties of Plastics.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Dept., 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI)

T-411 - Thickness (Caliper) of Paper and Paperboard.

(Copies of TAPPI Testing Methods may be obtained from the Technical Association of the Pulp and Paper Industry, One Dunwoody Park, Atlanta, GA 30341.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. Vulcanized fiber sheets (see 6.3) shall be commercial or trunk and case grade and shall be manufactured from cellulosic material treated with zinc chloride. The maximum use of reclaimed fiber is encouraged providing the finished material meets the requirements of this specification. The finished sheets shall have smooth calendered surfaces and shall contain not more than 0.10 percent zinc chloride.

3.2 Physical properties. The physical properties of the vulcanized fiber sheets, when tested as specified in 4.3.2, shall conform to the requirements specified in 3.2.1 through 3.2.6.

* 3.2.1 Density. The minimum density of sheets for the applicable thickness, as specified in the contract or order, shall be as specified in table I.

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TABLE I. Density requirements

Thickness inches	(mm)	Minimum density (g/cm^3)
0.010 through 0.015	(0.25 through 0.38)	1.10
0.020 through 0.094	(0.51 through 2.39)	1.15
0.100 through 0.625	(2.54 through 15.9)	1.20
0.688 through 1.00	(17.5 through 25.4)	1.10

3.2.2 Rockwell hardness. The Rockwell hardness of the vulcanized fiber sheet shall be R-50 minimum.

* 3.2.3 Bursting strength. The minimum bursting strength of sheets for the applicable thickness, as specified in the contract or order, shall be as specified in table II.

TABLE II. Bursting strength requirements

Thickness inches	(mm)	Minimum bursting strength, psi	(kPa)
0.010	(0.25)	125	860
0.012	(0.30)	145	1000
0.015	(0.38)	185	1280
0.020 and 0.025	(0.51 and 0.64)	250	1720
0.030 through 0.040	(0.76 through 1.02)	375	2590
0.047 through 0.055	(1.19 through 1.40)	560	3860
0.060	(1.52)	750	5170

* 3.2.4 Water absorption. The maximum water absorption of sheets for the applicable thickness, as specified in the contract or order, shall be as specified in table III.

TABLE III. Water absorption requirements

Thickness inches	(mm)	Maximum water absorption, percent 2 hour	24 hour
0.010 through 0.030	(0.25 through 0.76)	---	---
0.032 through 0.060	(0.81 through 1.52)	60	68
0.063 through 0.115	(1.60 through 2.92)	52	66
0.125 and 0.156	(3.2 and 4.0)	35	61
0.188 and 0.218	(4.8 and 5.5)	24	56
0.250 and 0.281	(6.4 and 7.1)	20	52
0.313	(8.0)	17	47
0.375	(9.5)	15	43
0.438	(11.1)	14	39
0.500 and 0.563	(12.7 and 14.3)	13	36
0.625 and 0.688	(15.9 and 17.5)	11	31
0.750 and 0.813	(19.1 and 20.7)	10	27
0.875 and 0.938	(22.2 and 23.8)	8	23
1.00	(25.4)	8	21

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* 3.2.5 Impact strength. For sheets 0.063 inch (1.60 mm) and above, the minimum impact strength, Izod, edgewise notched, shall be 1.20 foot pounds per inch (64.1 N·m/m) of notch in the cross direction and 1.60 foot pounds per inch (85.4 N·m/m) of notch in the machine direction.

* 3.2.6 Flexural strength. For sheets 0.063 inch (1.60 mm) and above, the minimum flatwise flexural strength shall be 13,000 psi (90 MPa) in the cross direction and 15,000 psi (103 MPa) in the machine direction.

* 3.3 Dimensions. Length, width, and thickness of sheets shall be as specified (see 6.2). A tolerance of ± 1/2 inch (± 13 mm) will be permitted on length and width dimensions. Thickness and thickness tolerances shall be as specified in table IV.

TABLE IV. Thickness requirements

Thickness inches	(mm)	Tolerance inches	(mm)
0.010 0.012 0.015	(0.25) (0.30) (0.38)	± 0.001	± 0.03
0.020	(0.51)	± 0.002	± 0.05
0.025 0.030 0.032	(0.64) (0.76) (0.81)	± 0.003	± 0.08
0.035 0.040 0.047	(0.89) (1.02) (1.19)	± 0.004	± 0.10
0.050 0.055 0.060 0.063	(1.27) (1.40) (1.52) (1.60)	± 0.005	± 0.13
0.065 0.070 0.075	(1.65) (1.78) (1.90)	± 0.007	± 0.18

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TABLE IV. Thickness requirements (cont'd)

Thickness inches	(mm)	Tolerance inches	(mm)
0.080	(2.03)		
0.085	(2.16)	± 0.007	± 0.18
0.094	(2.39)		
0.100	(2.54)		
0.115	(2.92)	± 0.009	± 0.23
0.125	(3.18)		
0.156	(3.96)		
0.188	(4.78)		
0.218	(5.54)		
0.250	(6.35)	± 0.012	± 0.30
0.281	(7.14)		
0.313	(7.95)		
0.375	(9.52)		
0.438	(11.13)		
0.500	(12.70)	± 0.015	± 0.38
0.563	(14.30)		
0.625	(15.88)		
0.688	(17.48)		
0.750	(19.05)	± 0.025	± 0.64
0.813	(20.65)		
0.875	(22.22)		
0.938	(23.83)	± 0.040	± 1.02
1.00	(25.40)		

3.4 Color. Unless otherwise specified by the contracting officer, the vulcanized fiber sheets shall be olive drab, approximating color 24087 of FED-STD-595. Color shall be produced by a blend of pigments incorporated with the fiber during manufacture.

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3.5 Marking of individual sheets. The applicable Federal Stock Number shall be placed on one side of each sheet in at least one place within 12 inches (305 mm) of a narrow edge and in numbers of not less than 1/2 inch (13 mm) in height by means of a pressure sensitive adhesive backed label.

3.6 Workmanship. The vulcanized fiber shall be clean and uniform in color and texture and shall conform to the quality and grade of product established by this specification.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. The end item shall be examined in accordance with the classification of defects at the inspection levels and acceptable quality levels (AQLs) set forth in 4.3.1.5. The lot size shall be expressed in units of sheets, of the size specified, for the examination in 4.3.1.1 and 4.3.1.2 and in units of shipping containers for the examination in 4.3.1.3.

4.3.1.1 Visual examination. The end item shall be examined on both sides for defects in appearance and workmanship. The sample unit for this examination shall be one sheet.

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Defects

Any hole, gouge, tear, cut, or foreign matter.
Any surface roughness or blister.
Any delamination.
Any cracking, checking, or rough edges.
Color not as specified.
Color not uniform, mottled, blotchy, or spotted.

4.3.1.2 Examination of the end item for dimensional defects. The sample unit for this examination shall be one sheet. Any length or width dimension that varies by more than ± 1/2 inch (13 mm) from that specified shall be classified as a defect.

4.3.1.3 Packaging inspection. An examination shall be made to determine that the packing and marking comply with section 5 requirements. The sample unit for this examination shall be one shipping container, fully prepared for delivery except that it need not be closed. Examination for closure defects listed below shall be made on shipping containers fully prepared for delivery.

Examine	Defect
Packing	Not level specified; not in accordance with contract requirements. Container, required methods, materials not as specifie Closure not accomplished as specified. Any component missing, damaged, or not as specified. Inadequate application of components such as incomplet closure of case liners and container flaps, improper taping and stapling, or loose or inadequate strapping Bulged or distorted containers.
Weight	Weight of contents exceeds requirements.
Marking	Omitted, incorrect, illegible, incomplete, or not as specified.

4.3.1.4 Palletization inspection. An examination shall be made to determi that the palletization complies with the section 5 requirements.. Defects sh be scored in accordance with the list below. The sample unit shall be one palletized unit load fully prepared for delivery. The lot size shall be the number of palletized unit loads in the end item inspection lot.

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Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement.
Palletization	Not as specified. Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

4.3.1.5 Inspection levels AQLs. The inspection levels and the AQLs expressed in defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQLs
4.3.1.1	S-4	4.0
4.3.1.2	S-2	4.0
4.3.1.3	S-2	2.5
4.3.1.4	S-1	6.5

4.3.2 Testing of the end item. The physical values specified apply to the average of determinations made on a sample unit for test purposes. The lot size shall be expressed in units of sheets. The sample unit shall be one sheet. Not more than one sample unit shall be drawn from any one shipping container. The inspection level for determining the sample size shall be S-1. The lot shall be unacceptable if one or more units fail to meet any requirement specified in table V.

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TABLE V. Instructions for testing

Characteristic	Specification reference: Requirement	Specification reference: Test method	No. determinations sample per/unit	Results reported as Numerically to nearest 2/
Zinc chloride content	3.1	ASTM D 619	2	0.01 percent
Density	3.2.1	ASTM D 619	2	0.01 g/cm^3
Rockwell hardness 1/	3.2.2	ASTM D 785 Procedure A	5	R-1
Bursting strength 1/	3.2.3	ASTM D 202	4 each side	5 psi (10 kPa)
Water absorption: 1/:				
2 Hours	3.2.4	ASTM D 570 Procedure 6.2	3	1 percent
24 Hours	3.2.4	ASTM D 570 Procedure 6.1	3	1 percent
Impact strength: 1/				
Crosswise	3.2.5	ASTM D 256 Method A	5	0.01 ft. lbs/in (0.1 N·m/m)
Lengthwise	3.2.5	ASTM D 256 Method A	5	0.01 ft. lbs/in (0.1 N·m/m)

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TABLE V. Instructions for testing (cont'd)

Characteristic	Specification reference: Requirement	Specification reference: Test method	No. determinations sample per/unit	Results reported as Numerically to nearest 2/
Flexural strength: 1/				
Crosswise	3.2.6	ASTM D 790	5	100 psi (1 MPa)
Lengthwise	3.2.6	ASTM D 790	5	100 psi (1 MPa)
Thickness 1/	3.3	4.4.2	5	0.001 inch (0.01 mm)

1/ Test specimens shall be conditioned in accordance with ASTM D 619 except as altered by the specific ASTM test procedure.

2/ Test reports shall include all values on which results are based.

4.4 Test methods.

* 4.4.1 Conditioning. Conditioning shall be in accordance with Procedure A of ASTM D 618 except that the conditioning period prior to test shall be at least 48 hours for specimens 1/8 inch (3 mm) or less in thickness and 96 hours minimum for thicker specimens.

4.4.2 Thickness. Thickness of the vulcanized fiber shall be measured at random positions throughout the sheet and shall be determined in accordance with TAPPI Method T-411.

5. PACKAGING

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

* 5.1.1 Level A packing. Fiber sheets of one description only shall be packed in a snug-fitting shipping container conforming to style optional, grade V2s of PPP-B-636 or overseas type, style A or I, grade A or B, type 2 load of PPP-B-601, as specified (see 6.2). Each fiberboard container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of PPP-B-636. Each wood container shall be closed

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and reinforced in accordance with the appendix of PPP-B-601 and waterproofed with a sealed case liner conforming to type I or II, grade C of MIL-L-10547. The weight of the contents of each fiberboard container shall not exceed 65 pounds (29.5 kg) and for each wood container shall not exceed 150 pounds (68.0 kg). Inspection shall be in accordance with 4.3.1.3.

* 5.1.2 Level B packing. Fiber sheets of one description only shall be packed in a snug-fitting shipping container conforming to style optional, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636 or domestic type, style A or I, grade B, type 2 load of PPP-B-601, as specified (see 6.2). Each fiberboard container shall be closed in accordance with method II as specified in the appendix of PPP-B-636. Each wood container shall be closed and reinforced in accordance with the appendix of the container specification. The weight of the contents of each fiberboard container shall not exceed 65 pounds (29.5 kg) and for each wood container shall not exceed 150 pounds (68.0 kg). Inspection shall be in accordance with 4.3.1.3.

* 5.1.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the fiberboard shipping container specified in 5.1.2 shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III of PPP-B-636 except that the inspection shall be in accordance with 4.3.1.3.

* 5.1.3 Level C. Fiber sheets shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.2 Palletization. Unless otherwise specified (see 6.2), fiber sheets of one description only, packed as specified in 5.1, shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

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5.3 Marking. In addition to any special marking required by the contract, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The vulcanized fiber sheet is intended for use in the fabrication of locker trunks, field desks, shipping cases for typewriters, and filing racks.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Dimensions and thickness required (see 3.3 and 6.4).
(c) Color required, if other than olive drab (see 3.4 and 6.4).
(d) Selection of the applicable levels of packing (see 5.1).
(e) Type of shipping container desired for level A or B packing (see 5.1.1 and 5.1.2).
(f) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.1.2.1).
(g) When palletization is not required (see 5.2).

6.3 Commercial comparison. The vulcanized fiber covered by this specification is similar to the grades and quality of commercial and trunk and case fiber described in publication VU1 of the National Electrical Manufacturers Association, NEMA Standards.

* 6.4 Available sizes and colors. Vulcanized fiber sheets are available in widths ranging from 47 to 53 inches (1194 to 1346 mm), and in lengths from 80 to 90 inches (2032 to 2286 mm) and in the following standard colors: red, gray, black, green, walnut, mahogany, granite, and brown (see 3.3 and 3.4).

* 6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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* 6.6 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

Custodians:

Army - GL
Air Force - 99

Review activities:

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User activities:

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